PROBLEM STATEMENT

Smart Hospital Management System Using Doubly Linked List and Queue in C++

Problem Overview

Managing patient registration, maintaining medical visit records, handling waiting queues, and tracking patient discharge manually can be time-consuming and inefficient. It may lead to delays in patient processing, difficulty retrieving patient information, and challenges in monitoring hospital operations.

The objective of this project is to develop a **Smart Hospital Management System using C++** that simplifies patient registration, queue management, and discharge tracking. The system uses a **Doubly Linked List** to maintain patient records and support forward and reverse traversal, while a **FIFO Queue** ensures that patients are called in their registration order.

The system provides automatic token generation, patient detail entry, predefined department selection, patient search by token number, display of patient records, waiting queue counting, estimated waiting time, and a hospital dashboard. It also includes a **patient discharge option** that updates the status of a consulted patient to “Discharged” while preserving the patient's record for future reference.

The main aim of this project is to improve patient record organization, simplify hospital queue handling, track patient status from registration to discharge, and demonstrate the practical application of doubly linked lists, queues, pointers, and dynamic memory allocation in C++.